Appendix: Discreteness Arises Not from an Outer Bound but from Nodes

— Discreteness as Nodes Generated by Binarity

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Purpose

This appendix develops independently §3 ("Derivation of discreteness") of the related paper. It adds no new claim; its aim is to make explicit an easily misread point — that **discreteness arises intrinsically not from an outer bound (boundedness) but from the nodes that Axiom 2 (binarity) generates**.

The explanation "discreteness comes from boundedness" implicitly borrows the eigen-oscillation of a string (only an integer number of half-waves fit on a string fixed at both ends). But an end (boundary) is a privileged point with a neighbor only on one side, and violates anonymity (the lemma of the parent paper). In this system, where ends are forbidden by anonymity, one cannot borrow the boundary condition of a string. Stating "discrete because bounded" nonetheless smuggles in the string's boundary condition — this is circular.

This appendix avoids this circularity and shows that discreteness comes intrinsically from Axiom 2 (binarity). The root of discreteness is not the outer bound (boundedness) but the nodes that binarity necessarily generates.

This appendix is not a rigorous proof but an explication of the structure of §3 of the parent paper. No physical identification is made.

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1. Axiom 2 generates nodes

Axiom 2 of this system (the limit of anonymity = 2) folds the interior distinction into a binary — "no other" and "an other present" (parent paper §0.5, §2). That is, what stands in the interior is binary ($+/-$, being/non-being).

When a quantity takes binary values ($+/-$), to move from the $+$ region to the $-$ region it must necessarily pass through the boundary of the two. This boundary — the point where the sign switches from $+$ to $-$ (or $-$ to $+$) — is called a **node**.

Being binary itself generates nodes. As long as a $+$ region and a $-$ region exist, there is necessarily a boundary (a node) between them. This arises not by imposing a boundary condition from outside, but is generated intrinsically by binarity (Axiom 2). For a continuous quantity, a sign switch is merely a single point passed through smoothly; but under binarity (Axiom 2), the switch from $+$ to $-$ is carved as a node.

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2. Nodes are discrete

A node is a point — the point of a particular position, the boundary between $+$ and $-$. A point is a discrete object, not a continuum.

Therefore, the existence of nodes immediately means the existence of discrete objects. Between nodes there is a $+$ or $-$ region, and the nodes themselves line up discretely. Discreteness follows directly from the existence of nodes as discrete points.

This requires no outer bound (the boundedness of the size of the system). Nodes arise intrinsically from binarity (Axiom 2), and from the fact that those nodes are discrete points, discreteness follows. There is no need to impose an upper bound on the size of the system.

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3. Anonymity binds the nodes to an integer count

A node is a point (a $+/-$ switch). By anonymity, no node has privilege — all nodes are equal (the anonymity of the parent paper).

When equal nodes line up, the number of nodes must be an integer. A node is binary — "present" or "not present" (Axiom 2) — and a fractional node (1.5 nodes) cannot exist. Nodes are discrete objects counted one, two, …, and their total is an integer.

That is, anonymity binds the nodes to an integer count. That the number of nodes is an integer is a consequence of the nodes being equal (anonymous) and binary (present/not present).

[keepaspectratio]fig5$_{n}odes_{o}n_{c}losed_{l}oop.pdf$

図 1 Figure 1: Nodes on a closed loop. Binary (+/−) values alternate, and their switch points (nodes) arise. Nodes are discrete points (hence discrete), bound to an integer count by anonymity, and to an even count by closure (the sign returns to + after one round). Discreteness comes not from an outer bound but from the nodes that binarity generates.

4. A closed system binds the nodes to an even count

The system is closed (it has no ends and returns upon going around once; the lemma of the parent paper). When binary (+/−) values alternate on a closed system, to return to the original sign (+) after one round, the sign switch must occur an even number of times: $+ \to - \to + \to$ ….

Therefore, on a closed system the number of nodes is even. The requirement of closure — that the sign returns after one round — binds the number of nodes to be even.

Since the interval between nodes corresponds to a half-wavelength, the number of nodes being even corresponds to the admissible wavelengths (modes) being numbered by integers (parent paper §3, stage two, λ_n=L/n).

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5. The structure of the derivation

From the above, discreteness comes intrinsically not from an outer bound (boundedness) but from the nodes that binarity (Axiom 2) generates.

Stage	Ground	Result
Genesis of nodes	Axiom 2 (binarity, +/− switch)	nodes arise
Discreteness	nodes are points (discrete)	discrete objects exist
Integrality	anonymity (nodes equal and binary)	the node count is an integer
Evenness	closure (sign returns after one round)	the node count is even

The root of discreteness is the nodes that binarity (Axiom 2) generates. An outer bound (an upper bound on the size of the system) is not needed for discreteness. On a closed system (the lemma of the parent paper, a self-unprovable premise), binarity carving the nodes into an integer / even count makes the admissible wavelengths (modes) numbered by integers (Figure 1).

Avoidance of circularity: "discrete because bounded" was a circularity borrowing the string's boundary condition. The derivation of this appendix does not borrow a string but derives the nodes intrinsically from Axiom 2 (binarity). Discreteness is a consequence of binarity — an intrinsic structure — not of the boundedness of the size of the system.

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6. Limitations (reading notes)

This appendix makes explicit the structure of §3 of the parent paper and adds no new claim.

1. **The closed system is a self-unprovable axiom (lemma).** There is a line "anonymity forbids ends → closes" (the lemma of the parent paper), but it is not fully derived from a more fundamental principle. The evenness (§4) of this appendix presupposes this closure.
2. **Discreteness comes from Axiom 2 and anonymity.** This appendix separates discreteness from the outer bound (boundedness) and attributes it to the nodes that Axiom 2 (binarity) generates. However, the derivation of Axiom 2 itself is outside the scope of the parent paper and this appendix.
3. **"Node" refers to the switch point of the interior binary.** No identification with a physical wave node is made. It is limited to the formal structure of the switch of the binary (+/−, being/non-being).
4. **Boundedness is unnecessary for discreteness but is separately involved in the integrality of mode numbers.** §3, stage two, of the parent paper states that the standing wave on a closed system (circumference L) is numbered by integer modes λ_n=L/n. The evenness of nodes (§4) of this appendix corresponds to this. The boundedness (the size of the system) itself is not the ground of discreteness, but the circumference L of the closed system is involved in the numbering of modes.
5. **No physical identification is made.** The physical names of the binary, nodes, and modes are anonymous labels, and this appendix makes no identification with a particular physical quantity.

Reference

- Kihara, N. "On the Logarithmic Map of Scale and the Separation of Scale and Quantization in a High-Symmetry Limit — A Limited Verification Report from a Minimal Axiom System," §3, the lemma of §2 (closure is a consequence of anonymity), Axiom 2. Concept DOI: 10.5281/zenodo.20740841.